

LANGKAWI INTL, Malaysia

WMO: 486000

Lat: 6.3367N

Lon: 99.7264E

Elev: 29

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	74.8	75.1	64.8	91.8	84.2	66.5	97.6	83.7	14.4	81.3	12.0	84.0	3.8	60	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	12.9	93.6	78.4	92.8	78.7	91.5	79.3	83.7	89.3	82.8	88.5	82.5	88.1	9.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.2	168.1	88.7	81.0	161.1	87.7	80.8	160.0	87.5	47.8	89.3	46.8	89.8	46.3	88.8	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	11.9	10.3	DB	71.9	96.3	1.0	1.7	71.2	97.6	70.6	98.6	70.0	99.5	69.3	100.8	(4)
(5)			WB	68.1	85.2	1.7	1.4	66.9	86.3	65.9	87.1	64.9	87.9	63.7	89.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	83.6	84.7	84.8	84.4	83.8	83.5	82.6	82.6	82.2	82.3	82.8	83.1
	DBStd	2.02	1.97	1.96	2.13	1.74	1.62	1.73	1.70	1.82	1.80	1.63	1.80	1.91
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12172	1040	971	1078	1032	1048	1004	1011	1010	965	1001	985	1027
	CDD65	6697	575	551	613	582	583	554	546	545	515	536	535	562
	CDH74	77321	6539	6663	7553	7254	7081	6573	6162	6139	5589	5726	5789	6253
	CDH80	29762	2540	2946	3368	3147	2850	2565	2196	2209	1857	1874	2006	2204
Wind	WSAvg	4.3	5.8	6.1	4.9	3.9	3.6	3.8	3.5	3.9	3.7	3.6	4.0	5.4
Precipitation	PrecAvg	98.3	1.5	1.5	2.4	6.9	11.6	11.3	10.0	10.9	16.5	15.8	7.8	2.9
	PrecMax	127.9	5.6	7.4	6.4	12.6	18.8	15.5	16.3	19.1	33.3	26.3	17.7	6.7
	PrecMin	73.1	0.0	0.0	0.1	0.3	4.3	5.3	4.9	5.9	6.4	9.1	2.6	0.0
	PrecStd	14.1	1.7	2.0	1.9	3.5	4.6	3.3	3.9	4.2	7.2	5.2	3.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	95.3	96.4	93.6	91.8	91.6	90.0	90.0	90.0	91.5	91.8	91.8
		MCWB	78.3	77.1	78.3	81.0	82.2	82.3	81.1	80.9	81.2	80.1	80.1	79.0
	2%	DB	91.7	93.5	93.6	91.8	91.1	90.0	89.7	89.7	89.3	89.6	91.1	91.0
		MCWB	77.3	77.1	78.0	81.4	82.1	81.7	81.0	80.9	80.9	80.1	80.0	78.7
	5%	DB	91.0	91.8	91.7	91.2	89.8	89.7	89.0	88.2	88.0	88.1	89.5	89.6
		MCWB	77.1	77.2	79.1	81.3	81.6	81.5	80.9	80.6	80.6	80.3	79.8	78.6
	10%	DB	89.4	91.1	90.0	89.7	89.2	88.2	87.9	87.8	87.5	87.5	87.9	88.1
		MCWB	77.1	77.2	79.4	80.9	81.4	81.1	80.8	80.5	80.3	80.2	79.9	78.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	82.1	83.5	84.4	84.3	84.5	82.8	83.5	82.7	82.7	82.7	82.2
		MCDB	89.1	89.9	90.9	90.6	89.8	89.1	88.0	88.4	88.3	87.8	88.7	88.6
	2%	WB	80.4	80.9	82.2	83.3	83.3	83.3	82.3	82.3	81.9	82.0	81.7	80.9
		MCDB	87.7	88.9	89.4	89.5	88.6	88.3	87.8	87.5	87.3	87.2	87.9	87.8
	5%	WB	79.6	80.2	81.3	82.5	82.6	82.5	81.5	81.4	80.9	80.9	80.8	80.1
		MCDB	86.8	88.2	88.5	88.6	87.9	87.8	87.1	86.9	86.4	86.3	87.1	86.8
	10%	WB	78.7	79.5	80.6	81.8	82.0	81.7	80.8	80.7	80.4	80.3	80.1	79.4
		MCDB	85.9	87.6	87.8	87.8	87.3	87.1	86.5	86.4	85.7	85.6	86.1	86.0
Mean Daily Temperature Range	5% DB	MDBR	12.0	13.8	12.9	11.2	10.2	10.4	10.3	10.1	10.0	10.3	11.0	10.6
		MCDBR	13.6	14.8	14.5	12.8	11.1	11.0	11.3	10.7	10.7	11.4	12.6	12.2
		MCWBR	5.6	6.2	6.2	5.9	5.8	5.9	6.1	5.8	5.7	5.6	5.6	5.4
	5% WB	MCDBR	12.1	13.4	12.7	11.3	10.4	10.6	10.7	10.4	10.2	10.8	11.6	11.2
		MCWBR	6.3	6.8	6.6	6.2	6.0	6.2	6.4	6.2	5.9	5.9	5.8	5.7
Clear-Sky Solar Irradiance	taub	0.446	0.465	0.474	0.474	0.453	0.462	0.465	0.465	0.456	0.453	0.446	0.445	
	taud	2.376	2.291	2.297	2.307	2.390	2.382	2.359	2.343	2.366	2.382	2.395	2.394	
	Ebn at Noon	277	274	272	267	267	261	261	266	272	274	275	275	
	Edh at Noon	39	44	44	43	38	38	39	41	40	39	38	38	
All-Sky Solar Radiation	RadAvg	1682	1937	1911	1825	1599	1560	1527	1545	1494	1493	1480	1477	
	RadStd	117	105	161	93	101	89	78	69	69	120	98	127	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.40	+0.25	N/A	+0.54	+0.41	+0.34	N/A	N/A	+169	+169	

Nomenclature: See separate page